

Julie Yeh

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Professional Summary

Scientific Computing and Applied Mathematics graduate with experience in Python, R, statistical modeling, numerical methods, and AI/NLP coursework. Skilled in translating quantitative problems into structured analyses, building data-driven models, and communicating technical results clearly. Interested in applying computational and analytical skills to real-world problems in data science, AI, operations, and technology.

Skills

- **Artificial Intelligence / Machine Learning:** Natural Language Processing (word embeddings, sequence models, transformers, dependency parsing), PyTorch
- **Statistical Modeling & Data Analysis:** Regression, ANOVA, predictive analytics, statistical modeling, data visualization
- **Programming Languages:** Python, MATLAB, R, Fortran, SQL
- **Computational & AI-Assisted Development:** Numerical methods, high-performance computing, parallel processing, Git/Github, ChatGPT, Codex

Education

Stanford Online (Certification)

May 2025 - August 2025

Artificial Intelligence and Machine Learning for Language Applications (XCS224N): Natural Language Processing with Deep Learning

- **Coursework:** Word Embeddings (Word2Vec, SVD), Transformer Architectures & Attention, RNN/LSTM, Machine Translation (Seq2Seq), Pretraining & Fine-tuning, Question Answering Systems

University of California, Santa Cruz

Jan 2023 - Dec 2023

Masters of Science, Scientific Computing and Applied Mathematics

- **GPA:** 3.57

- **Coursework:** Numerical Linear Algebra, Classical/Bayesian Inference, Introduction to High Performance Computing

University of California, Santa Cruz

Sept 2019 - Dec 2022

Bachelor of Arts, Computational Mathematics (Honors)

- **GPA:** 3.68

- **Coursework:** Programming for Mathematics (Python), Linear Algebra

Professional Experience

Quantum Universal Education Website

May 2025

Open Source Contributor

- Automated site deployment by integrating a GitHub Actions workflow, enabling continuous deployment via GitHub Pages.
- Improved scalability and maintainability by developing reusable SCSS variables for typography and design consistency.
- Enhanced UI customization through new brand color support while collaborating in an open-source environment using Git, Jekyll, and Liquid templating.

Galileo Learning

June 2021 - July 2022

Camp Operations Intern

- Supported daily camp operations for 20+ campers by managing check-ins, preparing activity materials, and maintaining organized program logistics.
- Supervised camper groups to foster safe, engaging, and positive educational experiences.
- Developed leadership, communication, and problem-solving skills by coordinating with staff to ensure smooth daily operations and positive camper experiences.

Projects

R - Predictive Analysis of Property Sale Prices

Sep 2023 - Dec 2023

Graduate Statistical Analysis Project

- Analyzed 2,000+ housing records to identify key structural, neighborhood, and categorical factors significantly influencing property sale prices using exploratory data analysis, ANOVA, and regression modeling.
- Built predictive linear and logistic regression models that revealed major price drivers and improved

understanding of complex market valuation patterns.

- Applied rigorous diagnostic and validation techniques to produce reliable, interpretable insights, demonstrating how statistical modeling supports scalable data-driven forecasting and decision-making.

Natural Language Processing & Deep Learning Projects

May 2025 - August 2025

- Implemented machine learning and NLP models including word embeddings, neural dependency parsing, sequence-to-sequence translation with attention, and transformer-based architectures to solve diverse language processing tasks.

- Developed hands-on experience with machine learning workflows, including model implementation, training, optimization, batching, masking, and performance evaluation using Python and PyTorch.

- Developed applied expertise in deep learning-based language systems, demonstrating how AI techniques can support language representation, prediction, and real-world machine learning applications.